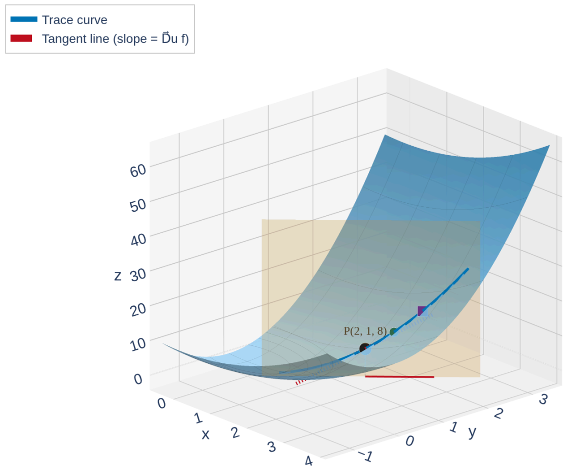
Section 14.6: Directional Derivatives and the Gradient Vector

Partial Derivatives

Directional Derivatives — Motivation

Partial derivatives give the rate of change along the **axes**. But what if we walk in some other direction?

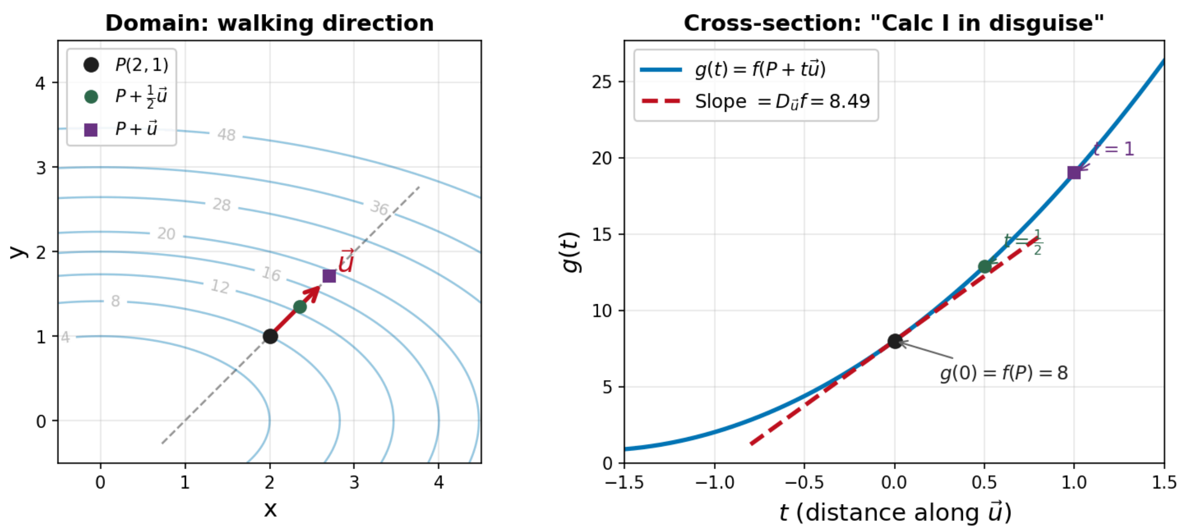
Consider $f(x, y) = x^2 + 4y^2$ at the point $P(2, 1)$. Pick a direction $\vec{u}$ at 45° and slice the surface with a vertical plane along that direction:



Interactive version

The vertical plane cuts a **trace curve** out of the surface. The slope of that trace at P is the rate of change in the $\vec{u}$ direction — the **directional derivative**.

From the side, this is just a single-variable function $g(t) = f(P + t\vec{u})$. Its derivative at $t = 0$ is the slope we want.



The directional derivative is just a **Calc I derivative in disguise**: pick a direction, slice the surface, and find the slope.

Definition of the Directional Derivative

f_x and f_y are just special cases of the directional derivative:

- $f_x = D_{\vec{i}}\, f$ — rate of change in the $\vec{i}$ direction
- $f_y = D_{\vec{j}}\, f$ — rate of change in the $\vec{j}$ direction

Directional Derivative

Let $\vec{u} = \langle a, b \rangle$ be a **unit vector** ($|\vec{u}| = 1$). The **directional derivative** of f at (x, y) in the direction of $\vec{u}$ is:

$$D_{\vec{u}} f(x, y) = \lim_{t \to 0} \frac{f(x + ta,\ y + tb) - f(x, y)}{t}$$

provided this limit exists.

Reading the formula:

- Start at (x, y)
- Move a tiny step t in the direction $\vec{u} = \langle a, b \rangle$, arriving at $(x + ta,\ y + tb)$
- Measure the change in f, divide by t, take the limit

This is the same **rise-over-run** idea from Calc I, but now we walk in the direction $\vec{u}$ instead of along the x-axis.

Example 1: $D_{\vec{u}} f$ from the Definition

Example 1

Let $f(x, y) = x^2 + 4y^2$ and $\vec{u} = \left\langle \frac{\sqrt{2}}{2},\ \frac{\sqrt{2}}{2} \right\rangle$.

Find $D_{\vec{u}} f(2, 1)$ using the **limit definition**.

The Gradient Vector

Before we find the shortcut, we need a new object.

Gradient Vector

If f has partial derivatives f_x and f_y, the **gradient** of f is the vector:

$$\nabla f(x, y) = \langle f_x(x, y),\ f_y(x, y) \rangle = f_x \vec{i} + f_y \vec{j}$$

The symbol ∇ is called **nabla** or **del**.

The gradient packages **both partial derivatives** into a single vector. It lives in the **domain** (the xy-plane), not on the surface.

Example 2

Find ∇f for $f(x, y) = x^2 + 4y^2$. Then evaluate $\nabla f(2, 1)$.

The Big Theorem: $D_{\vec{u}} f = \nabla f \cdot \vec{u}$

Theorem: Directional Derivative via the Gradient

If f is a differentiable function of x and y, then:

$$D_{\vec{u}} f(x, y) = \nabla f(x, y) \cdot \vec{u}$$

for any unit vector $\vec{u}$.

Translation: To find the rate of change in any direction, just take the **dot product** of the gradient with the direction vector.

This is the **central formula** of Section 14.6. It connects direction, rate of change, and gradient in one equation.

Why is this so much better than the definition?

- Limit definition: expand, simplify, take a limit (messy algebra)
- Gradient formula: compute two partial derivatives, take a dot product (fast!)

Why does it work? Recall the cross-section from our motivation: define $g(t) = f(x + ta,\ y + tb)$. Then by the Calc I derivative definition:

$$g'(0) = \lim_{t \to 0} \frac{g(t) - g(0)}{t} = \lim_{t \to 0} \frac{f(x + ta,\ y + tb) - f(x, y)}{t} = D_{\vec{u}} f(x, y)$$

So the directional derivative is just $g'(0)$. Now use the **chain rule** to find $g'(0)$ without limits.

Apply the **chain rule** (Section 14.5):

$$g'(t) = f_x(x + ta,\ y + tb) \cdot a\ +\ f_y(x + ta,\ y + tb) \cdot b$$

Evaluate at $t = 0$:

$$g'(0) = f_x(x, y) \cdot a + f_y(x, y) \cdot b = \langle f_x,\ f_y \rangle \cdot \langle a,\ b \rangle = \nabla f \cdot \vec{u}$$

The dot product appears **because of the chain rule**. Each partial derivative gets multiplied by the corresponding component of $\vec{u}$, which is exactly what a dot product does.

Example 3: $D_{\vec{u}}f$ via the Gradient

Example 3

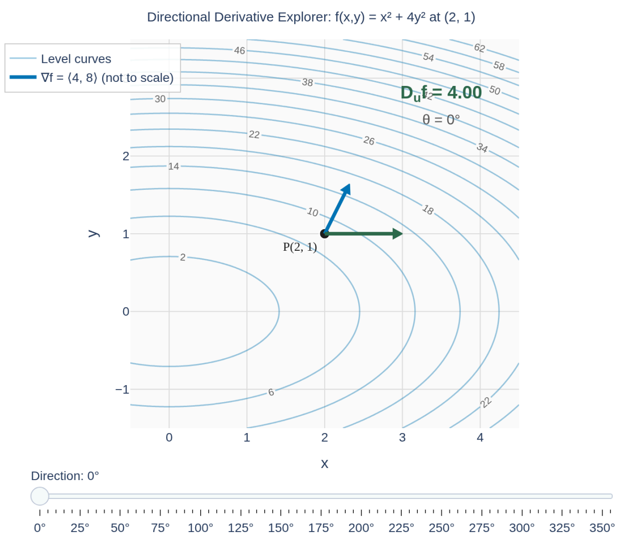
Same problem as Example 1: $f(x, y) = x^2 + 4y^2$, $\vec{u} = \left\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right\rangle$.

Find $D_{\vec{u}}f(2, 1)$ using the gradient.

Maximizing the Directional Derivative

Question: You're at $P(2, 1)$ on the surface $f(x, y) = x^2 + 4y^2$. Which direction increases f the fastest?

Use the slider to sweep $\vec{u}$ around and watch $D_{\vec{u}}f$ change:



Interactive version

Notice: $D_{\vec{u}}f$ is **largest** when $\vec{u}$ aligns with ∇f, **most negative** when opposing, and **zero** when perpendicular. Why?

Why? Rewrite $D_{\vec{u}}f$ using the dot product identity:

$$D_{\vec{u}}f = \nabla f \cdot \vec{u} = |\nabla f|\,|\vec{u}|\,\cos\theta = |\nabla f|\,\cos\theta$$

where θ is the angle between ∇f and $\vec{u}$.

$|\nabla f|$ is fixed at each point. The **only thing we control** is θ, and $\cos\theta$ ranges from -1 to 1:

θ	$\cos\theta$	$D_{\vec{u}}f$	Meaning
0	1	$\lvert\nabla f\rvert$ (max)	$\vec{u}$ **with** ∇f → steepest increase
$\pi/2$	0	0	$\vec{u} \perp \nabla f$ → along level curve
π	-1	$-\lvert\nabla f\rvert$ (min)	$\vec{u}$ **against** ∇f → steepest decrease

The gradient tells you everything:

- Its **direction** = steepest ascent
- Its **magnitude** $|\nabla f|$ = the maximum rate of change
- **Perpendicular** to it = no change (along the level curve)

Important: ∇f lives in the **domain** (the xy-plane), not on the surface. It points toward larger f-values in the input space.

Example 4: Fastest Increase and Decrease

Example 4

For $f(x, y) = x^2 + 4y^2$ at the point $(2, 1)$:

(a) What direction gives the **fastest increase** in f? How fast?

(b) What direction gives the **fastest decrease**?

(c) In what directions is $D_{\vec{u}} f = 0$?

Example 5: Which Way Should I Walk?

Example 5

A hiker is at position $(2, 1)$ on a mountain whose elevation is modeled by $f(x, y) = x^2 + 4y^2$ (hundreds of feet).

(a) She wants to **descend as steeply as possible**. What direction should she walk?

(b) She wants to **stay at the same elevation** (walk along a contour line). What direction(s) should she walk?

Your Turn: Directional Derivative Practice

Example 6

Let $f(x, y) = xe^{-y}$ and let $\vec{u}$ be the unit vector in the direction of $\langle 2, -1 \rangle$.

(a) Find $\nabla f(2, 0)$.

(b) Find $D_{\vec{u}} f(2, 0)$.

(c) What is the maximum rate of change of f at $(2, 0)$?

Gradient Is Perpendicular to Level Curves

Theorem: Gradient Perpendicular to Level Curves

The gradient vector $\nabla f(a, b)$ is **perpendicular** to the level curve $f(x, y) = k$ at the point (a, b).

Why does this make sense?

Along a level curve, f is constant, so the rate of change is **zero**. We saw that $D_{\vec{u}} f = 0$ when $\vec{u} \perp \nabla f$.

The tangent direction to a level curve is a direction of zero change — so it must be perpendicular to ∇f.

Look back at our explorer figure: $\nabla f(2, 1) = \langle 4, 8 \rangle$ visibly points **straight away from the ellipse** $f = 8$. Sweep the slider to $\theta \approx 153^\circ$ or 333° to see $D_{\vec{u}} f = 0$ — those are the tangent directions to the level curve.

The gradient always points **perpendicular to level curves**, aiming toward higher values of f. This is why contour maps and gradients are deeply connected.

Example 7: Tangent Line to a Level Curve

Example 7

Consider $f(x, y) = x^2 + 4y^2$ with level curves that are ellipses.
At the point $(2, 1)$, find the equation of the **tangent line** to the level curve $f(x, y) = 8$.

Extending to Three Variables

Everything generalizes naturally to functions of three variables $f(x, y, z)$:

Gradient in Three Variables

$$\nabla f(x, y, z) = \langle f_x,\ f_y,\ f_z \rangle$$

The directional derivative formula is exactly the same:

$$D_{\vec{u}} f(x, y, z) = \nabla f(x, y, z) \cdot \vec{u}$$

where $\vec{u} = \langle a, b, c \rangle$ is a unit vector in $\mathbb{R}^3$.

Level surfaces replace level curves:

For $F(x, y, z) = k$, the gradient ∇F is perpendicular to the **level surface** at each point.

Since ∇F is perpendicular to the level surface, it serves as the **normal vector** for the tangent plane!

Tangent plane to $F(x, y, z) = k$ at (a, b, c):

$$F_x(a, b, c)(x - a) + F_y(a, b, c)(y - b) + F_z(a, b, c)(z - c) = 0$$

Example 8: Tangent Plane to a Level Surface

Example 8

Let $F(x, y, z) = zx^2 - yz^2$.

(a) Compute $\nabla F(1, 2, 3)$.

(b) Find the equation of the tangent plane to the level surface $F(x, y, z) = -15$ at the point $(1, 2, 3)$.

Homework

Section 14.6: 1, 5, 9, 11, 13, 19, 21, 29, 31, 35, 39, 44, 49, 51, 55, 61, 65, 73